

KIRSCH, MI, USA

WMO: 725383

Lat: 41.817N

Lon: 85.433W

Elev: 282

StdP: 97.98

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 54827

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.6	-14.6	-21.3	0.6	-16.6	-18.0	0.8	-13.3	11.0	-2.6	9.5	-1.3	2.6	230	0.500

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	32.0	23.0	30.2	22.2	28.7	21.5	24.8	29.3	23.8	28.0	23.0	26.9	4.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	18.4	27.1	22.5	17.8	26.4	22.0	17.3	25.8	76.8	29.1	72.6	28.1	69.4	26.8	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.4	7.5	-21.8	33.9	3.7	1.7	-24.4	35.1	-26.6	36.1	-28.7	37.0	-31.4	38.3		
(5)			-22.0	26.4	3.6	1.1	-24.6	27.2	-26.7	27.8	-28.7	28.4	-31.3	29.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-4.2	-2.6	2.8	9.5	15.6	20.5	22.2	21.2	17.8	11.2	4.7	-1.2		
	DBStd	10.56	6.24	5.92	6.26	5.21	4.82	3.67	3.04	2.89	4.15	5.01	5.10	5.81		
	HDD10.0	1683	442	354	239	71	8	0	0	1	46	172	350			
	HDD18.3	3485	699	586	483	269	113	20	4	8	58	228	409	606		
	CDD10.0	1624	1	1	15	54	182	314	377	347	236	83	13	2		
	CDD18.3	385	0	0	1	3	28	84	123	97	43	7	0	0		
	CDH23.3	3106	0	0	4	29	243	691	1039	693	357	50	0	0		
	CDH26.7	870	0	0	0	3	57	207	326	169	100	8	0	0		
(14) Wind	WSAvg	3.2	3.8	3.7	3.7	4.0	3.5	2.8	2.2	2.1	2.4	3.1	3.6	3.7		
(15) Precipitation	PrecAvg	977	61	50	61	91	107	104	101	83	82	73	63			
	PrecMax	1268	135	127	132	185	188	188	196	269	188	197	138	137		
	PrecMin	721	23	16	18	10	36	28	56	36	26	17	20	19		
	PrecStd	124	31	29	25	41	45	41	34	45	41	43	28	27		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	16.0	23.4	27.0	31.1	32.7	33.8	32.4	32.2	27.6	20.5	15.6		
		MCWB	11.5	12.1	16.9	16.8	21.1	22.5	24.5	23.6	22.3	20.2	14.5	13.9		
	2%	DB	10.1	12.0	18.8	23.0	27.9	31.2	32.1	30.2	29.3	24.1	17.3	12.4		
		MCWB	8.7	9.8	14.2	15.1	18.9	22.0	24.0	22.7	21.3	17.7	13.1	11.1		
	5%	DB	7.1	7.8	15.9	21.2	26.3	28.9	30.1	28.0	27.4	22.2	14.8	9.0		
		MCWB	5.5	5.7	11.9	14.0	18.5	21.2	22.8	21.8	19.9	16.6	11.5	7.6		
	10%	DB	3.8	5.1	12.5	18.1	23.8	27.4	28.0	27.3	25.2	19.1	12.5	6.3		
		MCWB	2.5	3.0	9.1	12.2	17.5	20.5	21.8	21.3	18.8	15.2	9.5	4.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.3	13.3	17.6	18.9	22.7	25.0	26.5	25.4	24.1	21.6	16.0	14.1		
		MCDB	13.1	14.5	22.7	24.6	27.7	30.1	31.0	30.4	29.1	26.6	18.7	15.2		
	2%	WB	9.0	9.5	15.1	16.9	21.1	23.7	25.2	24.1	22.7	19.5	14.1	11.0		
		MCDB	9.5	11.5	18.2	21.1	26.0	28.2	29.6	28.1	27.3	22.7	16.6	12.1		
	5%	WB	5.5	6.0	12.4	15.1	19.9	22.7	24.2	23.3	21.5	17.6	11.9	7.7		
		MCDB	6.8	8.0	14.8	19.4	24.3	27.1	28.4	26.9	25.3	20.2	14.2	8.7		
	10%	WB	2.7	3.3	9.3	13.3	18.6	21.7	23.2	22.3	20.3	15.7	9.9	4.8		
		MCDB	3.7	4.9	12.2	17.9	22.4	25.8	26.9	25.7	23.8	18.9	12.3	6.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	7.7	9.1	10.9	10.7	10.3	10.6	10.3	11.2	10.0	7.9	6.6		
		MCDBR	9.1	10.5	12.8	13.9	12.8	12.0	11.9	11.4	13.0	12.9	10.8	9.3		
		MCWBR	8.3	8.5	8.7	8.0	6.8	5.3	5.3	5.6	6.1	7.1	7.9	8.1		
	5% WB	MCDBR	8.8	10.1	11.4	12.0	11.1	9.7	10.1	9.5	10.4	10.7	9.7	8.6		
		MCWBR	8.5	8.7	8.5	8.3	6.8	5.3	5.5	5.5	5.8	7.3	7.9	8.1		
(40) Clear-Sky Solar Irradiance	taub	0.298	0.343	0.352	0.394	0.427	0.439	0.434	0.441	0.398	0.351	0.319	0.300			
	taud	2.372	2.232	2.333	2.240	2.190	2.198	2.222	2.191	2.295	2.423	2.463	2.415			
	Ebn at Noon	847	849	892	875	851	839	839	820	832	837	820	813			
	Edh at Noon	82	110	113	133	144	143	138	138	116	91	75	73			
(44) All-Sky Solar Radiation	RadAvg	1.44	2.29	3.49	4.62	5.33	6.02	6.11	5.30	4.32	2.80	1.75	1.21			
	RadStd	0.14	0.33	0.34	0.42	0.51	0.39	0.30	0.28	0.37	0.28	0.25	0.14			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (56 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page